

**26** *Use of the Data Booklet is relevant to this question.*

Assume that the electronic configurations of the second row of the transition elements and their ions are similar to those of the first row of transition elements and their ions.

Molybdenum(IV) fluoride,  $\text{MoF}_4$ , is a green ionic solid.

What is the electronic configuration of the molybdenum ion in this compound?

- A**  $[\text{Kr}] 4\text{d}^2 5\text{s}^2$       **B**  $[\text{Kr}] 4\text{d}^2 5\text{s}^0$       **C**  $[\text{Kr}] 4\text{d}^4 5\text{s}^0$       **D**  $[\text{Kr}] 4\text{d}^0 5\text{s}^2$